

Test Report

No.SDHL2307012944HI

Date: Aug 03, 2023

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ZHEJIANG ZHUOWEI FLOORING CO., LTD.
NO.288TH, JIN GANG AVENUE, FUCUN TOWN, JINDONG ZONE,
JINHUA CITY, ZHEJIANG PROVINCE, CHINA

Sample Description : ECOLOGIC MGO FLOORING

As above test item and its relevant information regarding to the submission are provided and confirmed by the applicant. SGS is not liable to either the test item or its relevant information, in terms of the accuracy, suitability, reliability or/and integrity accordingly.

SGS Ref No. : SDFS2307004281FF
Sample Receiving Date : Jul 24, 2023
Test Performing Date : Jul 24, 2023 to Aug 03, 2023
Test Performed : Selected test(s) as requested by applicant

Test Result Summary

No.	Test(s) Requested	Result(s)	Comments
1	To determine the flame spread index (FSI) and smoke-developed index (SDI) of the sample's surface burning characteristics when it is subjected to the conditions of specified in ASTM E84-23 "Standard Test Method for Surface Burning Characteristics of Building Materials"	See attached sheet	/
For further details, please refer to the following page(s)			

Signed for and on behalf of
SGS-CSTC Standards Technical Services Co., Ltd. Shunde Branch



Peter Zhao
Authorized Signatory

scan to see the report



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I. TEST CONDUCTED

This test was conducted in accordance with ASTM E84-23 Standard Test Method for Surface Burning Characteristics of Building Materials.

II. INTRODUCTION

The method, designated as ASTM E84-23, "Standard Method of Test for Surface Burning Characteristics of Building Materials", is designed to determine the relative surface burning characteristics of materials under specific test conditions. Results are expressed in terms of flame spread index (FSI) and smoke developed index (SDI).

The purpose of this test method is to determine the relative burning behavior of the material by observing the flame spread along the specimen. Flame spread and smoke developed index are reported. However, there is not necessarily a relationship between these two measurements.

III. TEST PROCEDURE

The tunnel is preheated to 65.6°C (150°F), as measured by the floor-embedded thermocouple located 7.09m (23.25 ft) downstream of the burner ports, and allowed to cool to 40.6°C (105°F), as measured by the floor-embedded thermocouple located 3.96m (13 ft) from the burners. At this time the tunnel lid is raised and the test sample is placed along the ledges of the tunnel so as to form a continuous ceiling 7.32m (24 ft) long, 304.8mm (12 in) above the floor. The lid is then lowered into place.

Upon ignition of the gas burners, the flame spread distance is observed and recorded every 30 seconds. Flame spread distance versus time is plotted ignoring any flame front recessions. If the area under the curve (A) is less than or equal to 97.5 ft*min, $FSI = 0.515 \cdot A$; if greater, $FSI = 4900/(195-A)$.

The test results for smoke shall be plotted and the area under the curve shall be divided by the area under the curve for heptane, multiplied by 100, and rounded to the nearest multiple of five to establish a numerical smoke-developed index (SDI).

IV. CONDITIONING

Prior to testing, the sample was conditioned to a constant weight at a temperature of (23±2.8)°C (73.4±5)°F and at a relative humidity of (50±5)%.



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Sample details

Sample description	Flooring
Sample size	1220mmx200mm
Thickness	8.0mm

Exposed face: Wood grain surface

Mounting methods:

The specimen was supported during testing by 1/6' round metal rods placed at nominal 18 inch intervals across the width of the test chamber. Three samples form a row, total 6 rows.

Test results:

Flame Spread Index, FSI	Smoke-developed Index, SDI
5	35

RATING:

The National Fire Protection Association Life Safety Code 101, Chapter 10, Section 10.2.3 "Interior Wall and Ceiling Finish Classification", has a means of classifying materials with respect to Flame Spread and Smoke Developed when tested in accordance with ASTM E84, UL 723 "Method of Test of Surface Burning Characteristics of Building Materials".

International Building Code, Chapter 8, Interior Finishes, Section 803 "Wall and Ceiling Finishes", was classified in accordance with ASTM E84 or UL 723. Such interior finish materials shall be grouped in the following classes in accordance with their flame spread and smoke-developed indexes.



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The classifications are as follows:

Classification	Flame Spread Index, FSI	Smoke-developed Index, SDI
Class A	0-25	0-450
Class B	26-75	0-450
Class C	76-200	0-450

Since the tested sample received a Flame Spread Index 5 and a Smoke-developed Index 35, it would meet the requirements of Class A interior Wall & Ceiling Finish Category.

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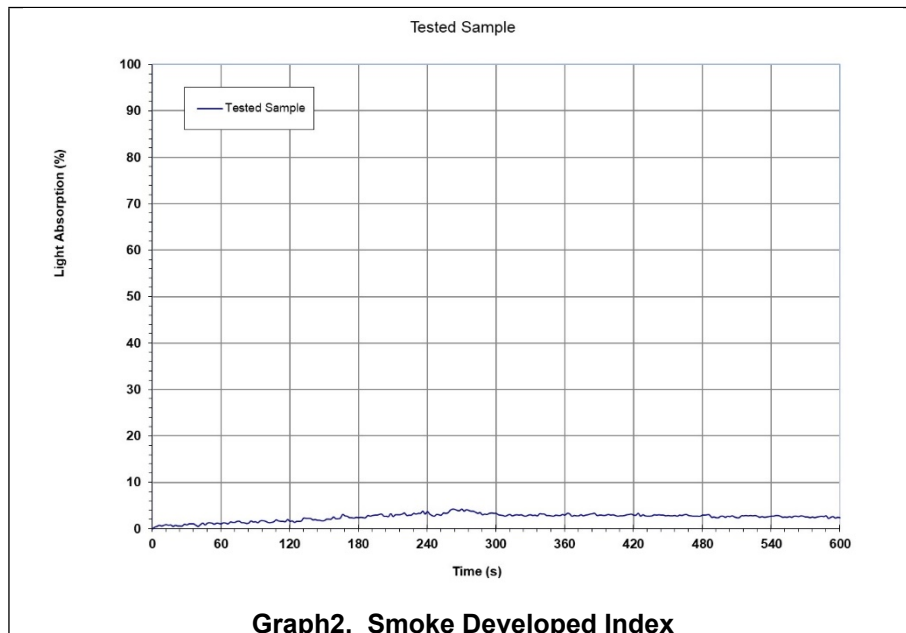
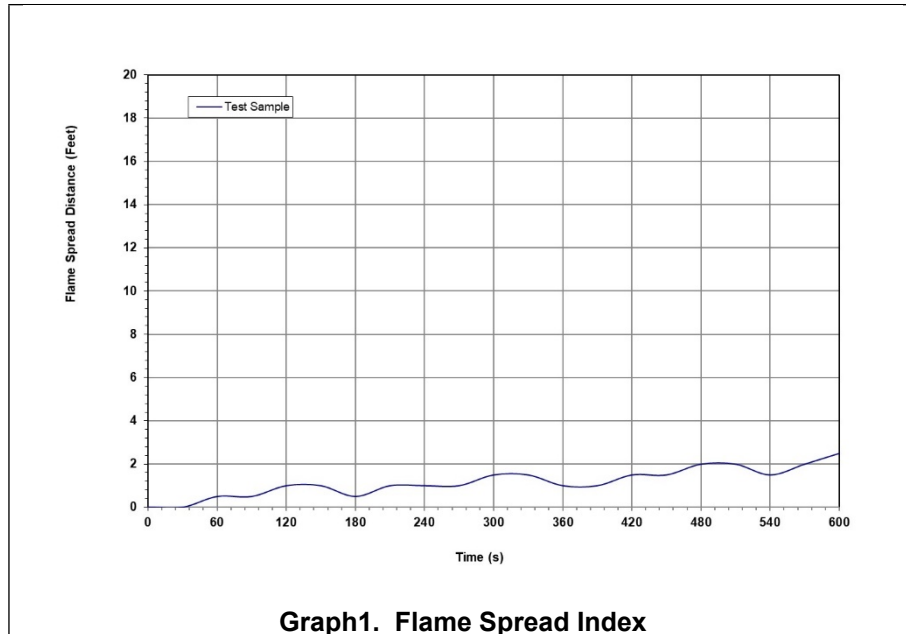


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GRAPHICAL RESULTS:



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OBSERVATIONS

Time to ignition (sec)	118
Time to Max. FS (sec)	590
Maximum FS (feet)	2.5
Observations	Non-flaming Debris

WARNING:

The use of supporting materials on the underside of the test specimen has the ability to lower the flame spread index from those which might be obtained if the specimen could be tested without such support. These test results do not necessarily relate to indices obtained by testing materials without such support.

Testing of materials that melt, drip, or delaminate to such a degree that the continuity of the flame front is destroyed, results in low flame spread indices that do not relate directly to indices obtained by testing materials that remain in place.

The test results relate only to the specimens of the product in the form in which were tested. Small differences in the composition or thickness of the product may significantly affect the performance during the test and may therefore invalidate the test results. Care should be taken to ensure that any product, which is supplied or used, is fully represented by the specimens, which were tested.

Statement:

This declaration of conformity is only based on the result of this laboratory activity, the impact of the uncertainty of the results was not included.

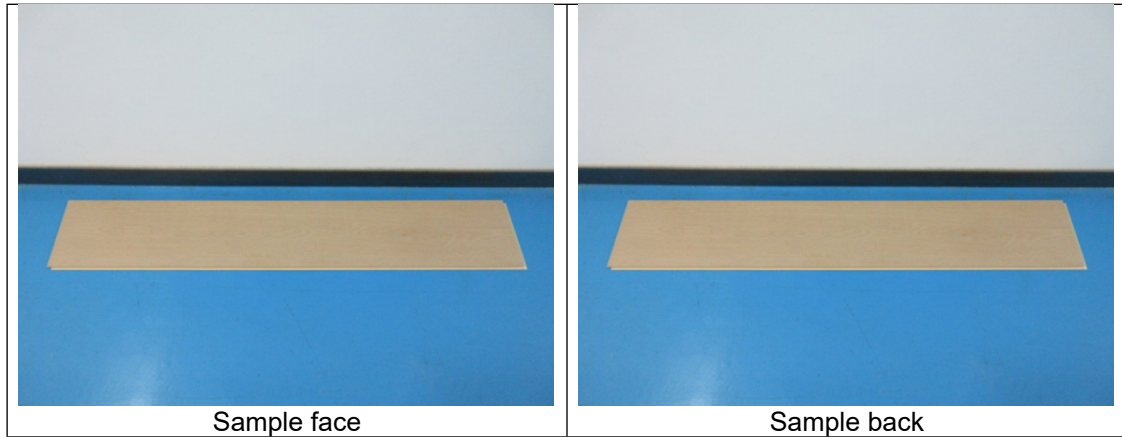


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Photo Appendix:



End of Report



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